

Technical Analysis System - Functional Description

Document Information

- **System:** Portfolio Tracker - Technical Analysis Module
 - **Version:** 1.0
 - **Date:** October 2025
 - **Purpose:** Functional specification of the technical analysis scoring system
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1. Executive Summary

The Technical Analysis System provides automated stock analysis using multiple technical indicators to generate buy/sell/hold signals. The system analyzes historical price data to calculate various indicators and combines them into a weighted scoring system that produces actionable trading signals.

2. System Overview

2.1 Purpose

The system helps investors make informed decisions by analyzing stock price movements through mathematical indicators that identify trends, momentum, and potential reversal points.

2.2 Key Features

- Multi-indicator technical analysis
 - Weighted scoring algorithm
 - Five-level signal classification
 - Automatic calculation from historical price data
 - Flexible data requirements (adapts to available data)
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3. Technical Indicators

The system calculates seven primary technical indicators, each serving a specific purpose in market analysis.

3.1 Simple Moving Average (SMA)

Purpose: Identifies price trends by smoothing out short-term fluctuations.

Calculation Periods:

- **SMA 20:** 20-day moving average (short-term trend)
- **SMA 50:** 50-day moving average (medium-term trend)
- **SMA 200:** 200-day moving average (long-term trend)

Interpretation:

- Price above SMA → Bullish signal (upward trend)
- Price below SMA → Bearish signal (downward trend)

Data Requirements:

- SMA 20: Minimum 20 price points

- SMA 50: Minimum 50 price points
 - SMA 200: Minimum 200 price points
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3.2 Exponential Moving Average (EMA)

Purpose: Similar to SMA but gives more weight to recent prices, making it more responsive to new information.

Calculation Periods:

- **EMA 12:** 12-day exponential moving average
- **EMA 26:** 26-day exponential moving average

Interpretation:

- Used primarily for MACD calculation
- More sensitive to recent price changes than SMA

Data Requirements:

- EMA 12: Minimum 12 price points
 - EMA 26: Minimum 26 price points
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3.3 Relative Strength Index (RSI)

Purpose: Measures the speed and magnitude of price changes to identify overbought or oversold conditions.

Calculation Period: 14 days (RSI 14)

Scale: 0 to 100

Interpretation:

- **RSI < 30:** Oversold condition → Strong buy signal
- **RSI > 70:** Overbought condition → Strong sell signal
- **RSI > 50:** Above neutral → Moderate bullish
- **RSI < 50:** Below neutral → Moderate bearish

Data Requirements: Minimum 15 price points

3.4 Moving Average Convergence Divergence (MACD)

Purpose: Shows the relationship between two exponential moving averages to identify trend changes and momentum.

Components:

- **MACD Line:** Difference between 12-day EMA and 26-day EMA
- **Signal Line:** 9-day EMA of the MACD line
- **Histogram:** Difference between MACD line and Signal line

Interpretation:

- **MACD > Signal:** Bullish momentum
- **MACD < Signal:** Bearish momentum
- **Histogram > 0:** Increasing bullish momentum

- **Histogram < 0:** Increasing bearish momentum

Data Requirements: Minimum 35 price points

3.5 Bollinger Bands

Purpose: Measures market volatility and identifies potential price extremes.

Components:

- **Middle Band:** 20-day SMA
- **Upper Band:** Middle + (2 × standard deviation)
- **Lower Band:** Middle - (2 × standard deviation)

Interpretation:

- Price near upper band → Potentially overbought
- Price near lower band → Potentially oversold
- Band width indicates volatility

Data Requirements: Minimum 20 price points

3.6 Golden Cross / Death Cross

Purpose: Identifies major trend reversals based on moving average crossovers.

Patterns:

- **Golden Cross:** SMA 50 crosses above SMA 200 → Strong bullish signal
- **Death Cross:** SMA 50 crosses below SMA 200 → Strong bearish signal

Interpretation:

- SMA 50 > SMA 200 → Long-term bullish trend
- SMA 50 < SMA 200 → Long-term bearish trend

Data Requirements: Both SMA 50 and SMA 200 must be available

4. Signal Generation Algorithm

4.1 Weighted Scoring System

The system uses a weighted points system where each indicator contributes points based on its reliability and importance.

Weight Distribution:

Indicator	Weight	Rationale
RSI 14	5	High reliability for momentum
Golden/Death Cross	4	Strong trend confirmation
MACD Signal	4	Reliable momentum indicator
SMA 20 vs Price	3	Short-term trend

SMA 50 vs Price	3	Medium-term trend
MACD Histogram	2	Additional momentum confirmation
SMA 200 vs Price	2	Long-term trend baseline

Total Maximum Weight: 23 points

4.2 Point Allocation Rules

Moving Averages (SMA 20, 50, 200):

- **Current Price > SMA:** Add weight to bullish points
- **Current Price < SMA:** Add weight to bearish points

RSI Analysis:

- **RSI < 30 (Oversold):** Add 5 points to bullish
- **RSI > 70 (Overbought):** Add 5 points to bearish
- **30 ≤ RSI ≤ 70 and RSI > 50:** Add 2.5 points to bullish
- **30 ≤ RSI ≤ 70 and RSI < 50:** Add 2.5 points to bearish

MACD Analysis:

- **MACD > Signal Line:** Add 4 points to bullish
- **MACD < Signal Line:** Add 4 points to bearish
- **Histogram > 0:** Add 2 points to bullish
- **Histogram < 0:** Add 2 points to bearish

Golden/Death Cross:

- **SMA 50 > SMA 200:** Add 4 points to bullish
 - **SMA 50 < SMA 200:** Add 4 points to bearish
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4.3 Net Score Calculation

Formula:

$$\text{Net Score} = (\text{Bullish Points} - \text{Bearish Points}) / \text{Total Weight}$$

Range: -1.0 to +1.0

Example Calculation:

- Bullish Points: 15
 - Bearish Points: 5
 - Total Weight: 20
 - Net Score = (15 - 5) / 20 = 0.5
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4.4 Signal Classification

Based on the Net Score, the system generates one of five signals:

Net Score Range	Signal	Meaning
≥ 0.6	STRONG BUY	Very bullish - multiple strong positive indicators

0.2 to 0.6	BUY	Moderately bullish - positive trend indicated
-0.2 to 0.2	HOLD	Neutral - no clear trend or mixed signals
-0.6 to -0.2	SELL	Moderately bearish - negative trend indicated
< -0.6	STRONG SELL	Very bearish - multiple strong negative indicators

5. Adaptive Data Requirements

5.1 Flexibility

The system adapts to available data, calculating only indicators for which sufficient data exists.

5.2 Minimum Requirements

- **Absolute Minimum:** 20 price points (allows basic SMA 20 and Bollinger Bands)
- **Recommended:** 200+ price points (enables all indicators)

5.3 Graceful Degradation

When insufficient data exists for certain indicators:

1. Those indicators are excluded from the calculation
 2. Total weight is adjusted accordingly
 3. Signal is generated from available indicators
 4. If no indicators can be calculated, system returns "HOLD"
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6. Example Scenario

6.1 Sample Stock Analysis

Input Data: 250 days of price history for Stock XYZ

- Current Price: \$150
- Price trend: Upward over past months

Calculated Indicators:

- SMA 20: \$145 → Price above (Bullish)
- SMA 50: \$140 → Price above (Bullish)
- SMA 200: \$135 → Price above (Bullish)
- RSI 14: 65 → Above 50, not overbought (Moderately Bullish)
- MACD: 2.5 (Signal: 1.8) → Above signal (Bullish)
- MACD Histogram: 0.7 → Positive (Bullish)
- Golden Cross: SMA 50 (140) > SMA 200 (135) → Active (Bullish)

Point Calculation:

- SMA 20 above: +3 bullish
- SMA 50 above: +3 bullish
- SMA 200 above: +2 bullish
- Golden Cross: +4 bullish
- RSI above 50: +2.5 bullish
- MACD above signal: +4 bullish

- MACD histogram positive: +2 bullish
- **Total Bullish: 20.5 points**
- **Total Bearish: 0 points**
- **Total Weight: 23 points**

Net Score: $(20.5 - 0) / 23 = 0.89$

Signal: STRONG BUY (Net Score ≥ 0.6)

7. Use Cases

7.1 Portfolio Review

Scenario: User wants to review all holdings

- System calculates technical signals for each position
- User can identify which stocks show strong buy/sell signals
- Helps in rebalancing decisions

7.2 New Investment Research

Scenario: User researching potential new stock purchase

- System analyzes technical indicators
- Provides objective signal based on price trends
- Complements fundamental analysis

7.3 Exit Strategy

Scenario: User holds profitable position

- System monitors for SELL or STRONG_SELL signals
 - Helps identify optimal exit timing
 - Prevents emotional decision-making
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8. Limitations and Considerations

8.1 Data Quality

- **Accuracy depends on input:** Garbage in, garbage out
- **Historical data only:** Cannot predict unexpected events
- **Market hours:** Only reflects traded prices

8.2 Not Financial Advice

- **Tool for analysis:** Not a recommendation to buy or sell
- **One perspective:** Should be combined with fundamental analysis
- **User responsibility:** Final investment decisions rest with user

8.3 Market Conditions

- **Works best in trending markets:** Less reliable in sideways markets
 - **Lagging indicators:** Based on past prices, not future predictions
 - **External factors:** Cannot account for news, earnings, or macro events
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9. System Benefits

9.1 Objectivity

- Removes emotional bias from analysis
- Consistent application of rules
- Quantifiable decision criteria

9.2 Efficiency

- Instant analysis of any stock
- Batch processing of entire portfolio
- Automated recalculation with new data

9.3 Educational Value

- Users learn technical analysis concepts
 - Understanding of market momentum
 - Pattern recognition development
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10. Future Enhancements

10.1 Potential Additions

- Volume-based indicators (OBV, VWAP)
- Fibonacci retracement levels
- Support and resistance levels
- Pattern recognition (head & shoulders, triangles)
- Candlestick pattern analysis

10.2 Customization Options

- Adjustable indicator periods
 - Custom weight assignments
 - User-defined signal thresholds
 - Indicator on/off toggles
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Appendix A: Glossary

Moving Average: Average price over a specified time period

Momentum: Rate of price change

Overbought: Price has risen too far too fast, correction likely

Oversold: Price has fallen too far too fast, bounce likely

Crossover: When one indicator crosses above/below another

Volatility: Degree of price variation over time

Trend: General direction of price movement

Signal Line: Moving average of another indicator (smoothing)

Appendix B: References

Technical Indicators Library: technicalindicators npm package

Calculation Methods: Standard industry formulas for all indicators

Best Practices: Based on common technical analysis principles used by traders worldwide

Document Version History

Version	Date	Changes
1.0	October 2025	Initial functional specification

End of Document